

Nashmi: Emergency Response Backend Architecture

Project Overview

Nashmi is a smart emergency response platform engineered to drastically reduce the critical time gap between incident occurrence and medical intervention. By establishing a direct, low-latency integration with national 911 dispatch centers, the system transforms verified nearby medical personnel into immediate first responders.

This project successfully qualified for the Crown Prince Award for Government Applications, demonstrating excellence in operational logic and digital service optimization.

System Architecture & Backend Logic

1. Dispatch Operations Engine (Central Control)

- Implemented real-time geographical tracking to monitor incident locations alongside available medical responders and critical supply points (e.g., pharmacies with AEDs).
- Engineered an automated dispatch algorithm that analyzes response time, geographical proximity, and equipment availability to ping the optimal subset of responders (limiting to 3 to prevent overlap).
- Developed auto-escalation logic: if a request goes unanswered within a strict timeframe, the system automatically redirects the ping to the next available tier.

2. Secure Identity Verification Pipeline

- Integrated backend verification processes linking directly to official government platforms (Sanad) to authenticate the medical credentials of first responders before granting system access.

3. Dual-Routing Navigation System

- Built dynamic, dual-destination routing logic that guides the responder to the incident while simultaneously calculating detours to the nearest medical support points for critical equipment acquisition.

Data Privacy & System Performance

- **Strict Data Minimization:** Designed the architecture to bypass personal patient data storage. The backend exclusively processes and transmits non-identifiable data packets consisting only of geographical coordinates and incident categorization.
- **Latency Optimization:** The current Minimum Viable Product (MVP) operates in the foreground to prove the viability of instantaneous, real-time data transfer. Future iterations are mapped to utilize dedicated background services for uninterrupted state management.

Core Technical Stack

- Cloud Backend Integration (Firebase)
- Algorithmic Dispatch Routing
- System Architecture & Data Minimization
- Real-time Geolocation Services